CV

AI CAD-to-Cost Analysis

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AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

46

Steering Knuckle RH

116 × 237 × 210 mm · 356.1 cm³ · Al 0.962 kg / Steel 2.795 kg

Manufacturability: 46/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	**116.5 × 236.8 × 210.1 mm**	Surface Area	**789.3 cm²**
Volume	**356.10 cm³**	Weight (Al)	**0.962 kg**
Weight (Steel)	**2.795 kg**	Weight (Plastic)	**0.374 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Bearing/hub bore	3	**High**	Large cylindrical bore family (r=31.5-44mm) for wheel-bearing seat
Mounting holes	14	**High**	14 drilled/reamed holes for ball-joint, tie-rod and caliper bracket mounting (r=5-7.5mm)
Threaded bores	1	**Medium**	Tapped holes for bolted joints
Boss/shaft features	3	**Medium**	Steering-arm and strut bosses (r=12.5-39mm)
Undercut/side-action faces	52	**High**	52 undercut faces requiring cores or side-pulls in the mould
Free-form blend surfaces	24	**Low**	B-spline fillets/blends typical of cast structural knuckle arms

§3 — Material Analysis

AI-suggested

Primary Material: **Ductile (nodular) cast iron GJS500-7 (72% confidence)**

Steering knuckles are safety-critical suspension components almost universally produced as ductile iron sand castings or forged steel, never thin-wall die-cast aluminium, due to fatigue/impact loading. The 52-undercut geometry, thick non-uniform walls (1.4-46.7mm), bosses and bearing bore all match a sand-cast ductile-iron knuckle machined on bore/hub/mounting faces. No aluminium signal (uniform 2-4mm HPDC wall, filename hint) is present, so per guidance the default for a solid drivetrain/suspension part is steel/ductile iron rather than aluminium.

Alternatives: **Forged/cast steel 1045 (18%)** · **Ductile iron GJS600-3 (higher strength) (8%)** · **Aluminium 6061 (lightweight alt., not chosen) (2%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sand cast + CNC machine	cast_and_machine	90%	0.2960	User-forced cast_and_machine; ductile-iron sand casting is the standard high-volume route for steering knuckles, with subsequent machining of bearing bore, mounting holes and mating faces.
Gravity die casting	casting	25%	0.1890	Feasible for aluminium alternative but not selected given steel/iron material choice.
Closed-die forging + machine	forging	20%	0.2960	Forged steel knuckles are an alternative production route offering higher fatigue strength but higher tooling/die cost; less favoured for the freeform cored geometry seen here.

§5 — Manufacturability Risks (Score: 46/100)

Severity	Feature / Area	Description	Recommended Action
High	**52 undercut faces**	Extensive undercuts require multiple side-cores/pulls in the sand mould, increasing pattern complexity and scrap risk from core shift.	Consolidate undercuts where possible or plan the pattern with pre-defined core prints and locating features.
High	**Min wall 1.36mm**	Very thin local sections adjacent to much thicker bosses (mean 23mm) risk misruns, cold shuts and porosity in sand casting.	Increase minimum wall to >4mm or redesign as a rib rather than a thin web; add chills/risers at thin-to-thick transitions.
Medium	**14 post-cast holes + threads**	Holes are cast-undersized or solid and finished by drilling/reaming/tapping in 3 setups, adding cycle time and requiring accurate as-cast datum locating.	Cast clean locating bosses/pads for fixturing and consider casting pilot holes to reduce drilling stock.
Medium	**Bearing bore (r=31.5-44mm)**	Bore is a critical, tightly toleranced feature that must be finish-bored/ground after casting; as-cast ovality and core shift will affect stock allowance uniformity.	Add generous machining allowance (2-3mm/side) and finish-bore in single setup off common datums.
Low	**Zero-draft faces (102)**	Large proportion of zero-draft planar faces increases pattern lift resistance and risk of sand tear-out.	Apply minimum 1-2° draft to non-machined faces where functionally acceptable.

§6 — DFM Issues (cast_and_machine)

Severity	Area	Description	Impact	Fix
High	**Undercuts / core-actions**	52 undercut faces demand multiple sand cores or side-pulls, raising pattern cost and risk of core shift causing wall-thickness variance.	Increases pattern tooling cost (~15-25%) and scrap rate (target <2% vs typical 5-8% with poor core control).	Re-orient part or simplify geometry to consolidate undercuts into fewer, larger cores with robust core prints.
High	**Thin web (1.36mm min)**	Local wall far below the 4-6mm sand-casting practical minimum for ductile iron risks incomplete fill and cold shut defects.	Casting scrap/rework, potential in-service fatigue failure risk if defect undetected.	Thicken minimum sections to "e4mm or convert thin web to a cast rib with generous fillets.
Medium	**Bearing bore finish machining**	Critical bore (Ø63-88mm equivalent) must be bored to tight tolerance from an as-cast surface with core-shift induced eccentricity.	Extra roughing pass or larger machining stock required, adding ~0.05-0.1hr/part machining time.	Add 2-3mm/side uniform machining allowance and use the bore itself as a locating datum in the casting pattern.

Severity	Area	Description	Impact	Fix
Medium	**Post-cast hole/thread finishing**	14 holes plus tapped features must be drilled/reamed/tapped after casting in up to 3 setups, adding non-value cycle time.	Adds ~7min/part drilling + setup overhead, ~10-15% of total machining cost.	Cast pilot/cored holes to near-finish size and combine hole operations into fewer setups using shared datums.
Low	**Zero-draft faces**	102 faces show zero draft, increasing pattern withdrawal force and sand tear risk.	Higher scrap/rework rate on non-critical cosmetic faces.	Add 1-2° draft on all non-machined faces during pattern design.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£19.50	**£27.00**	**£39.00**

Net Weight	**2.546 kg**	Material	**mat-gjs500**
Recommended Process	**cast_and_machine**	Cycle Time	**0.2161 hr/part**
Setup Time	**1.000 hr**	Operations	**4 ops**
Operations Detail	**Milling — +Z (60 faces) (mach-haas-vf2, 0.0184 hr) Milling — +X (51 faces) (mach-haas-vf2, 0.0156 hr) Milling — +Y (30 faces) (mach-haas-vf2, 0.0092 hr) Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured] (mach-drill, 0.1729 hr)**		

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£17,000**
Die Life	**8,000 shots**
Cavities	**2**
Yield	**65.0%**
Flash Weight	**0.300 kg**
Yield	**70.0%**
Die Cost	**£128,000**
Strokes	**6**
Cavities	**0**
Wall Thickness	**0 mm**

Mould Cost	**£0**
Mould Life	**0 shots**
Projected Area	**0.0 cm²**

§8 — AI Analysis Explanation

This is a right-hand steering knuckle, a safety-critical suspension component. Its sparse fill ratio (0.0615) combined with thick, non-uniform wall sections, 124 cylindrical bore/boss faces, 52 undercuts and a large bearing bore is characteristic of a sand-cast ductile-iron part machined on its functional interfaces (hub bore, ball-joint/tie-rod holes, caliper bracket faces). Per the forced cast_and_machine process and 200,000 units/year volume, a durable metal matchplate sand pattern (£17k, ~500k-shot life) feeding an automated green-sand ductile-iron foundry line is the standard high-volume route, followed by 3-setup CNC machining (5-axis for the undercut/hub faces, then drilling/tapping) totalling ~0.296hr cycle plus 0.75hr setup. Aluminium was deliberately not selected despite lower weight, because steering knuckles require the fatigue/impact toughness of ductile iron or forged steel and show none of the thin uniform HPDC wall signals that would indicate an aluminium die-cast design.

§9 — Analysis Limitations & Assumptions

1. Material (ductile iron vs forged steel vs aluminium) inferred from part type and geometry only; no drawing/BOM material callout was provided.
2. Cavity count and pattern life for sand casting are rule-of-thumb estimates for 200k/yr automated foundry, not derived from geometry.
3. Cost range is a parametric estimate; actual costs depend on foundry yield, local labour/energy rates and casting quality achieved.

Stage-1 pre-selection: cast_and_machine (100%) ·